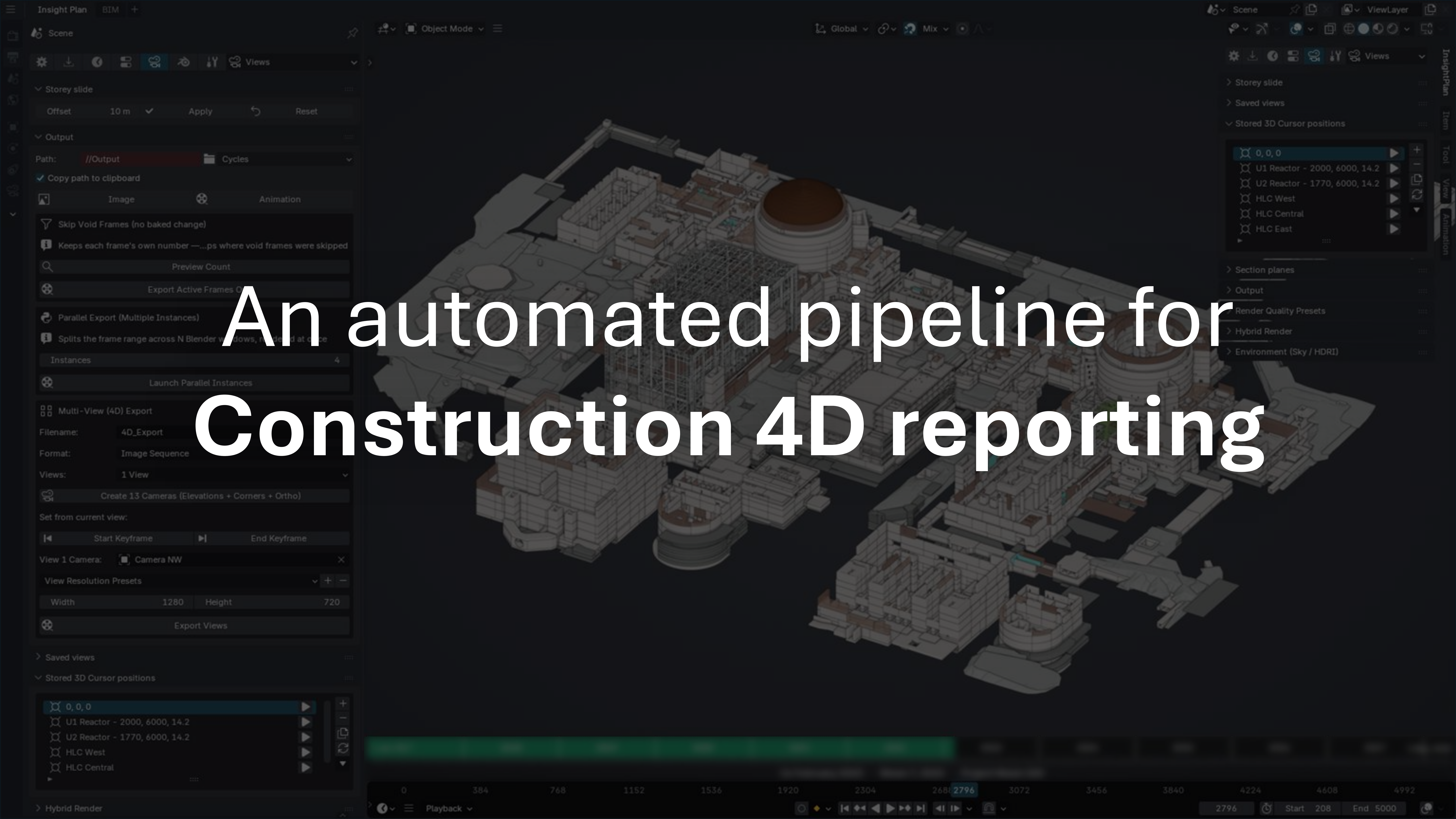


# An automated pipeline for Construction 4D reporting



# Speakers



**Hing Pang Huang**

**Developer (Pythoneer)**

at Hinkley Point C since November 2021



**David Felix**

**Digital Engineer and Planner**

at Hinkley Point C from 2017 to 2023  
currently at Sizewell C



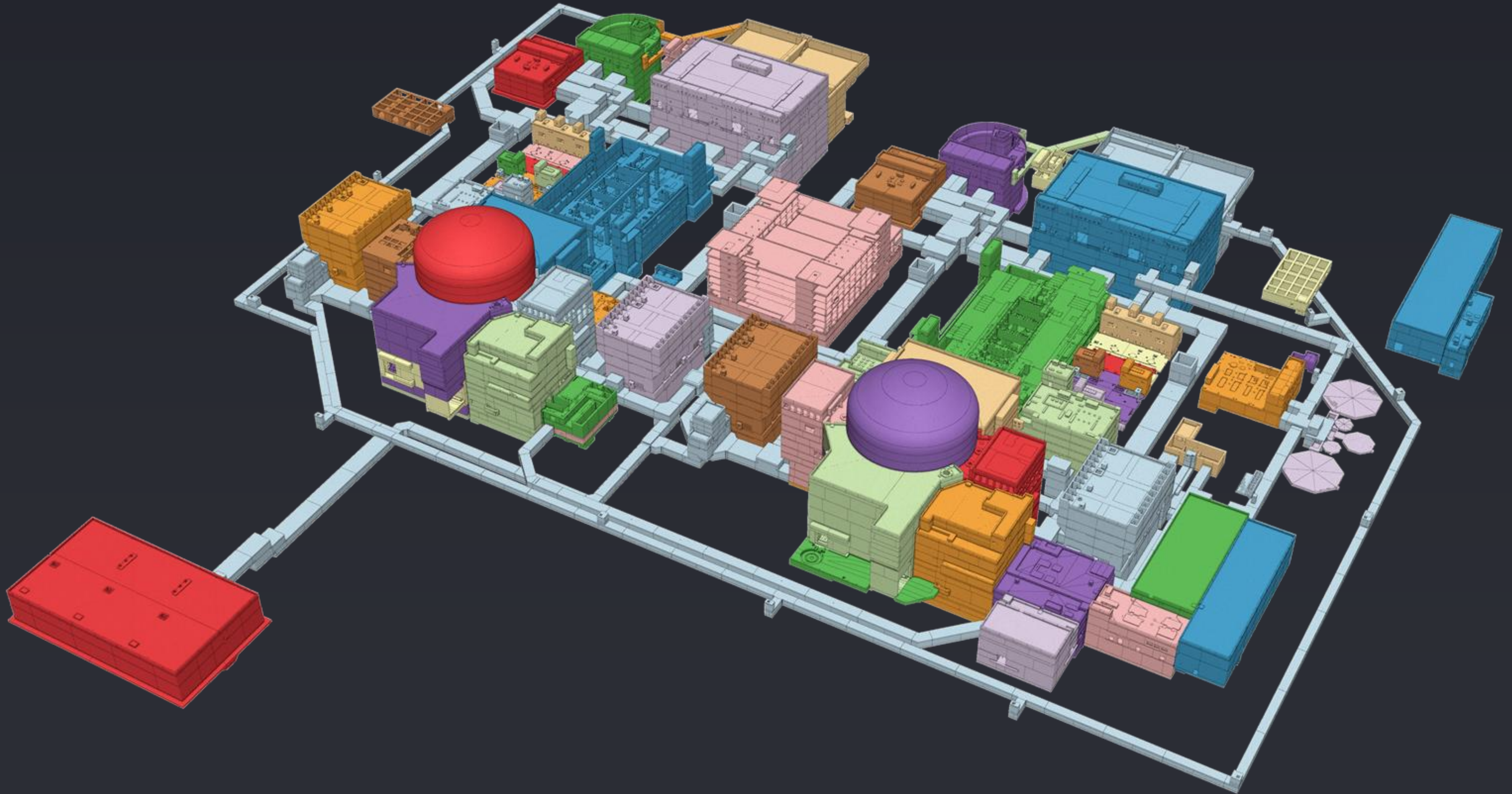
# Nuclear Power Station - Hinkley Point C



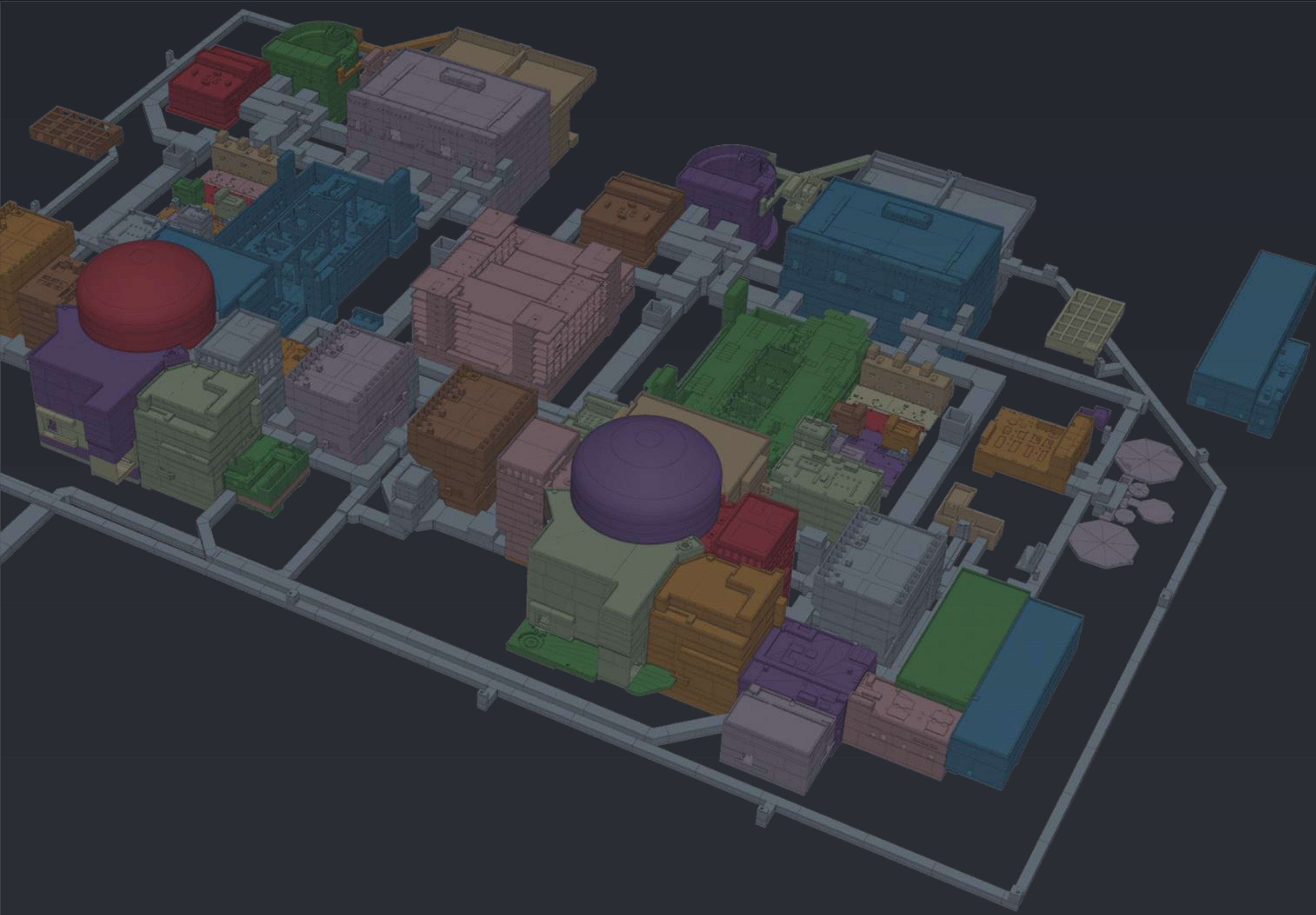
<https://www.youtube.com/watch?v=maWbEvM52Jo>

# Nuclear Power Station - Hinkley Point C

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# Nuclear Power Station - Hinkley Point C



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**50** Buildings

---

**55** Tower Cranes

---

**25** Planners

---

**400** Models of Concrete

---

**7200** Rooms

---

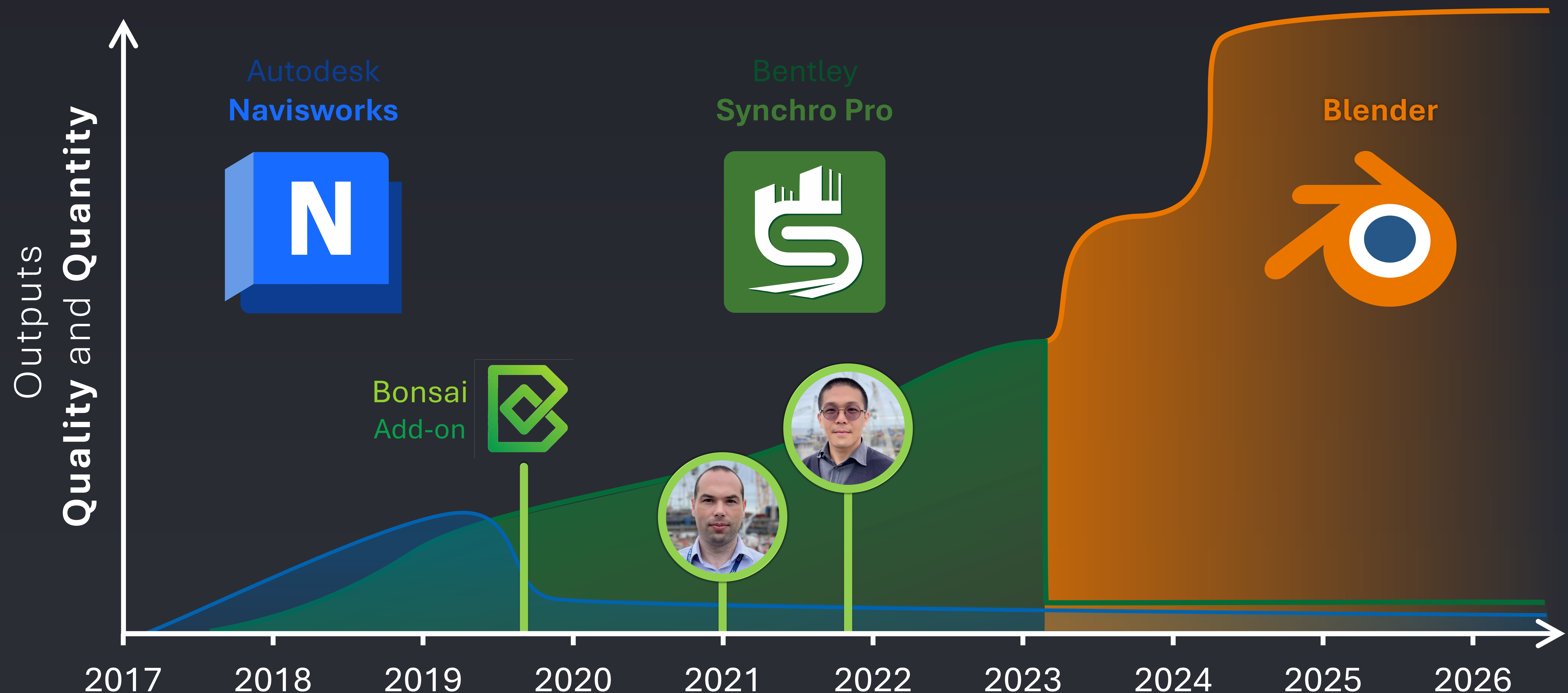
**35000** Concrete Pours

---

**1** Digital Engineer - 4D

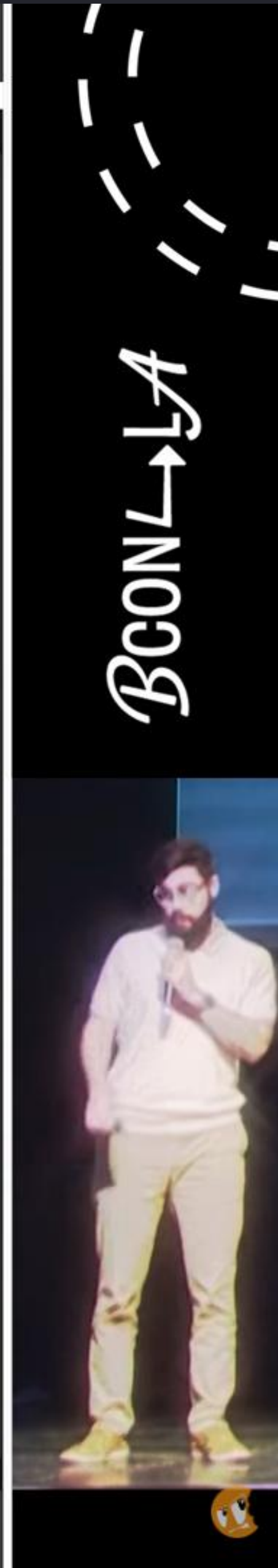
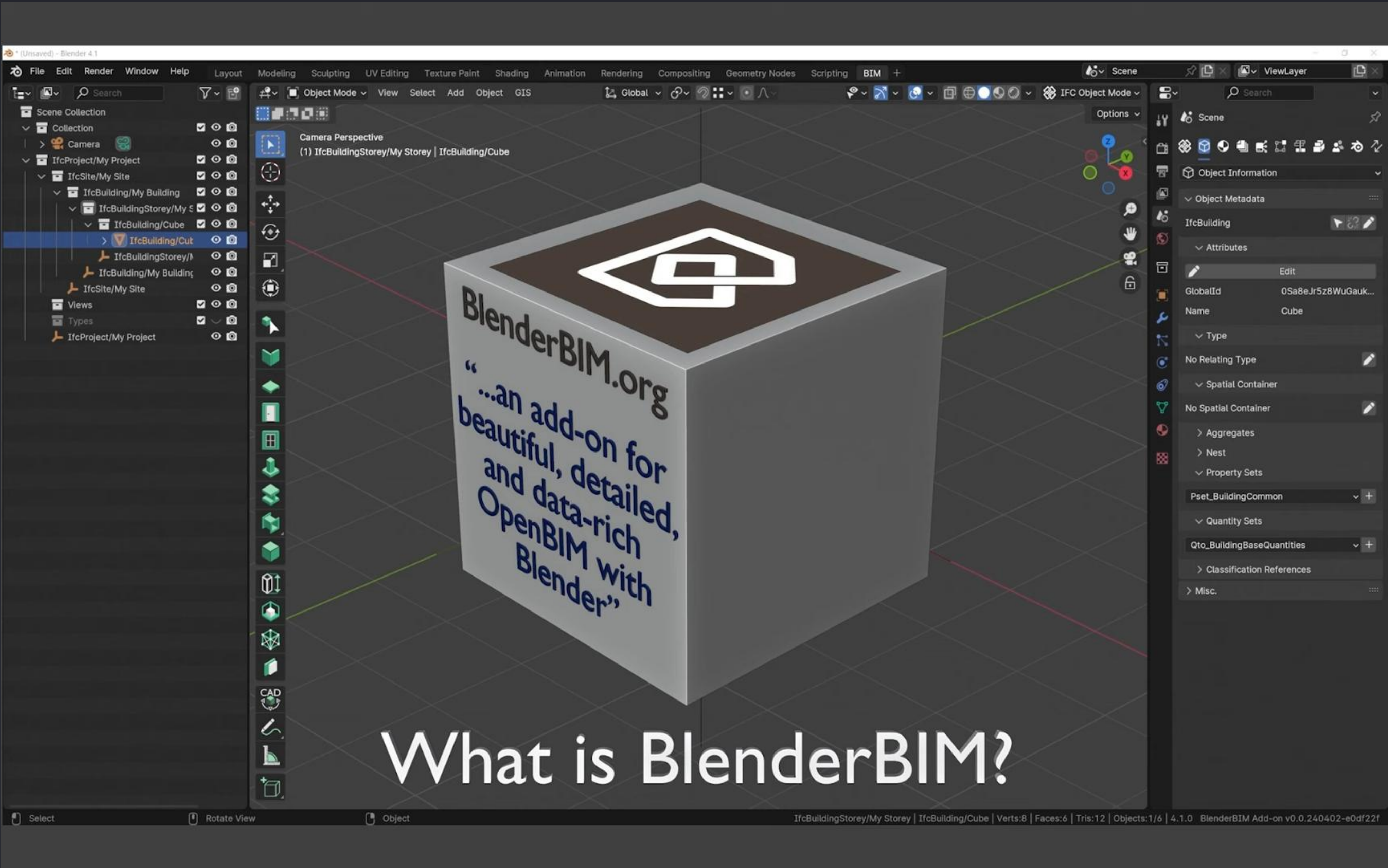
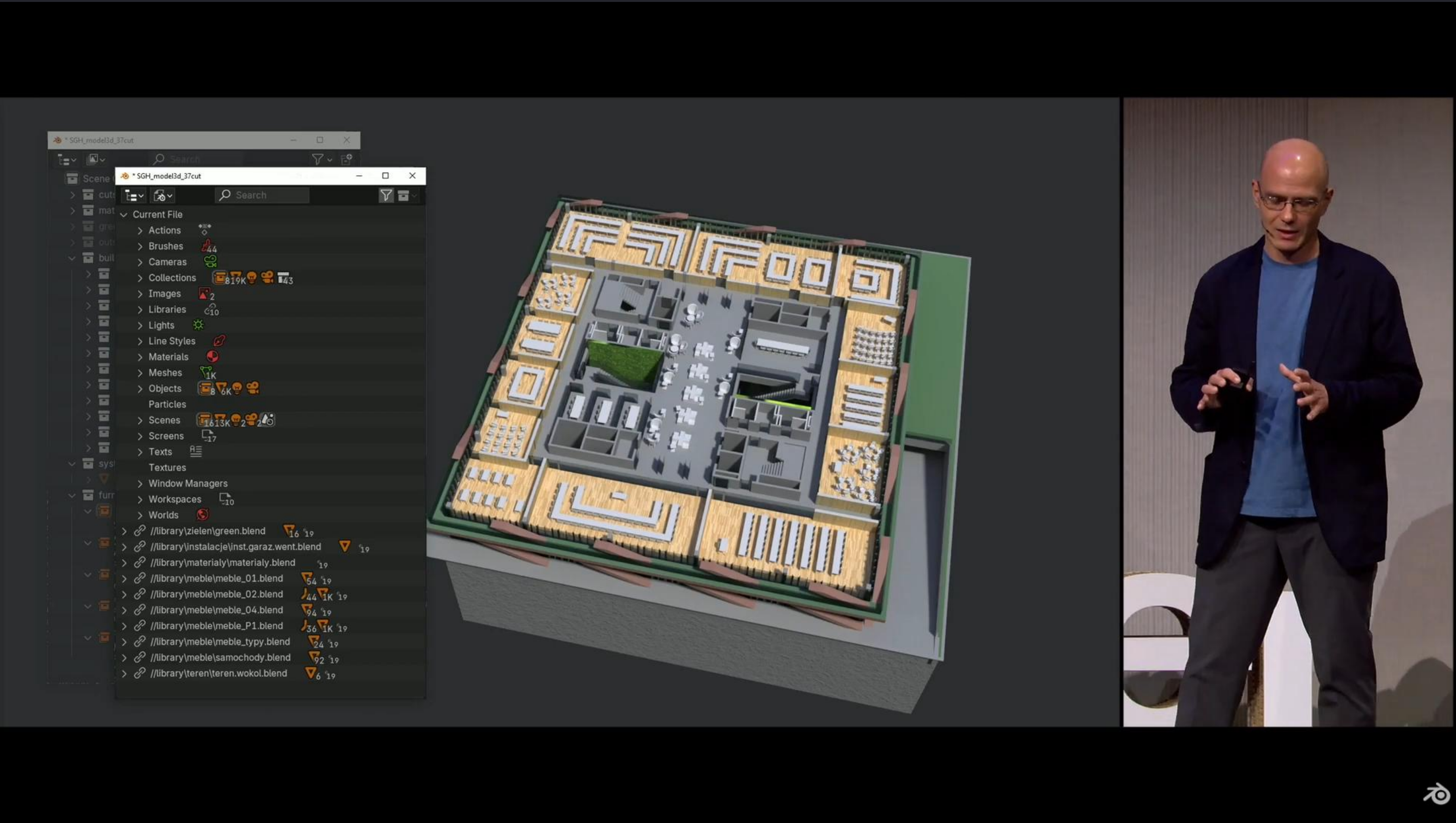
---

# Software used for 4D during the years



# Bonsai references

(previously BlenderBIM Add-on)



Blending in architectural design and on construction site  
by Paweł Czernecki



3D in Construction  
by Daniel Bridges & Renzo Caroca

# 3 key principles for successful 4D delivery

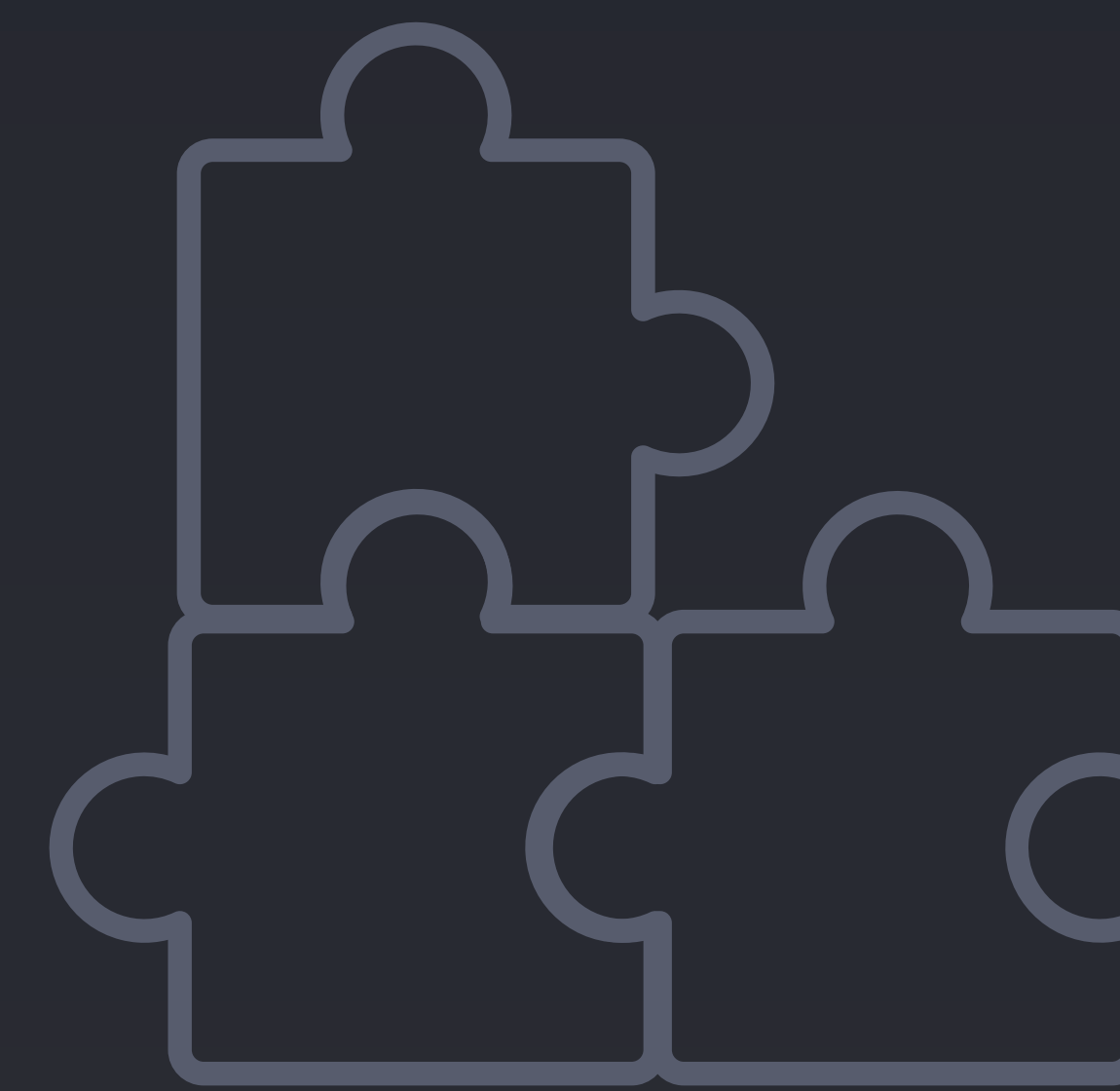
1

Start from the Output



2

Preparation is key

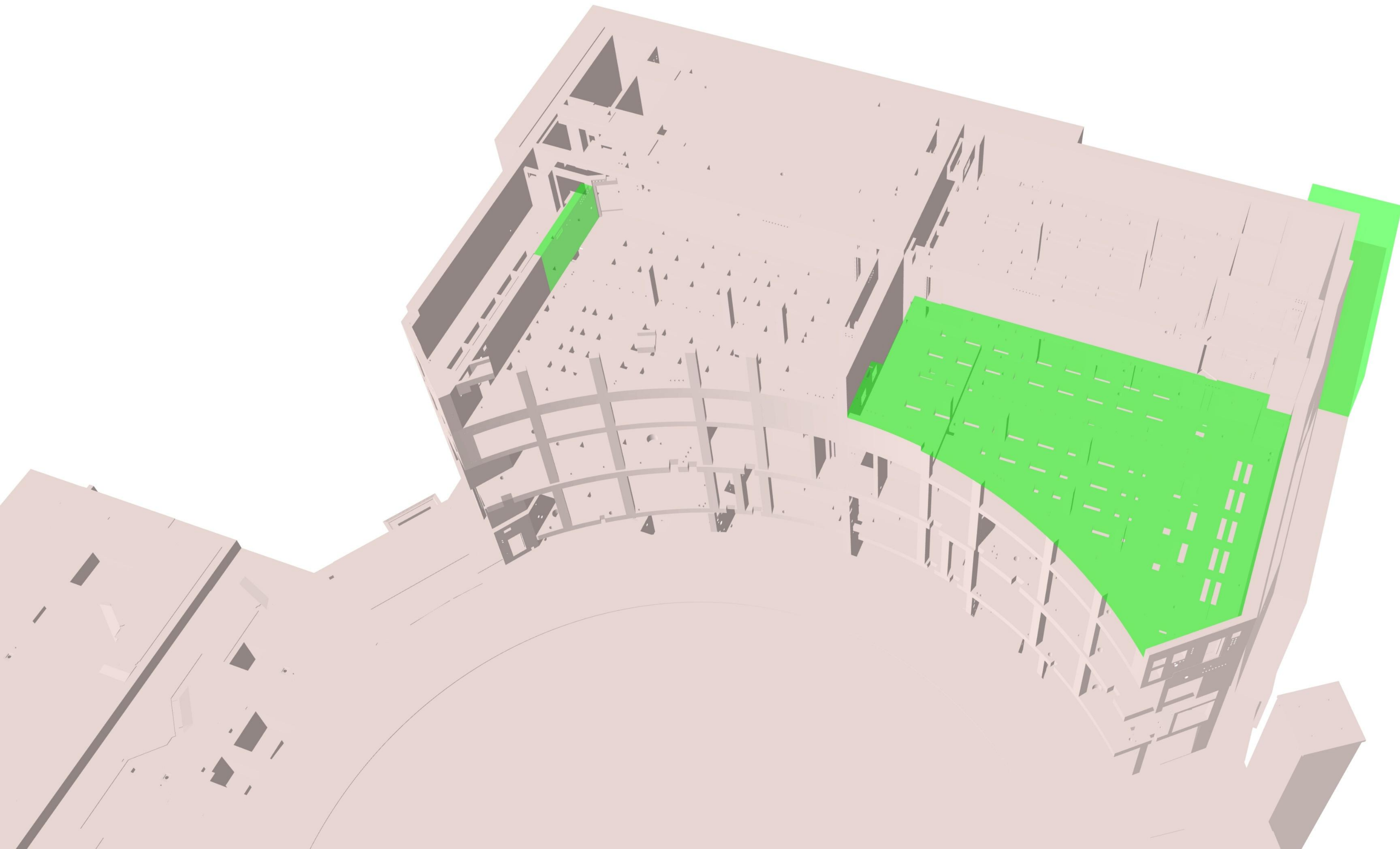


3

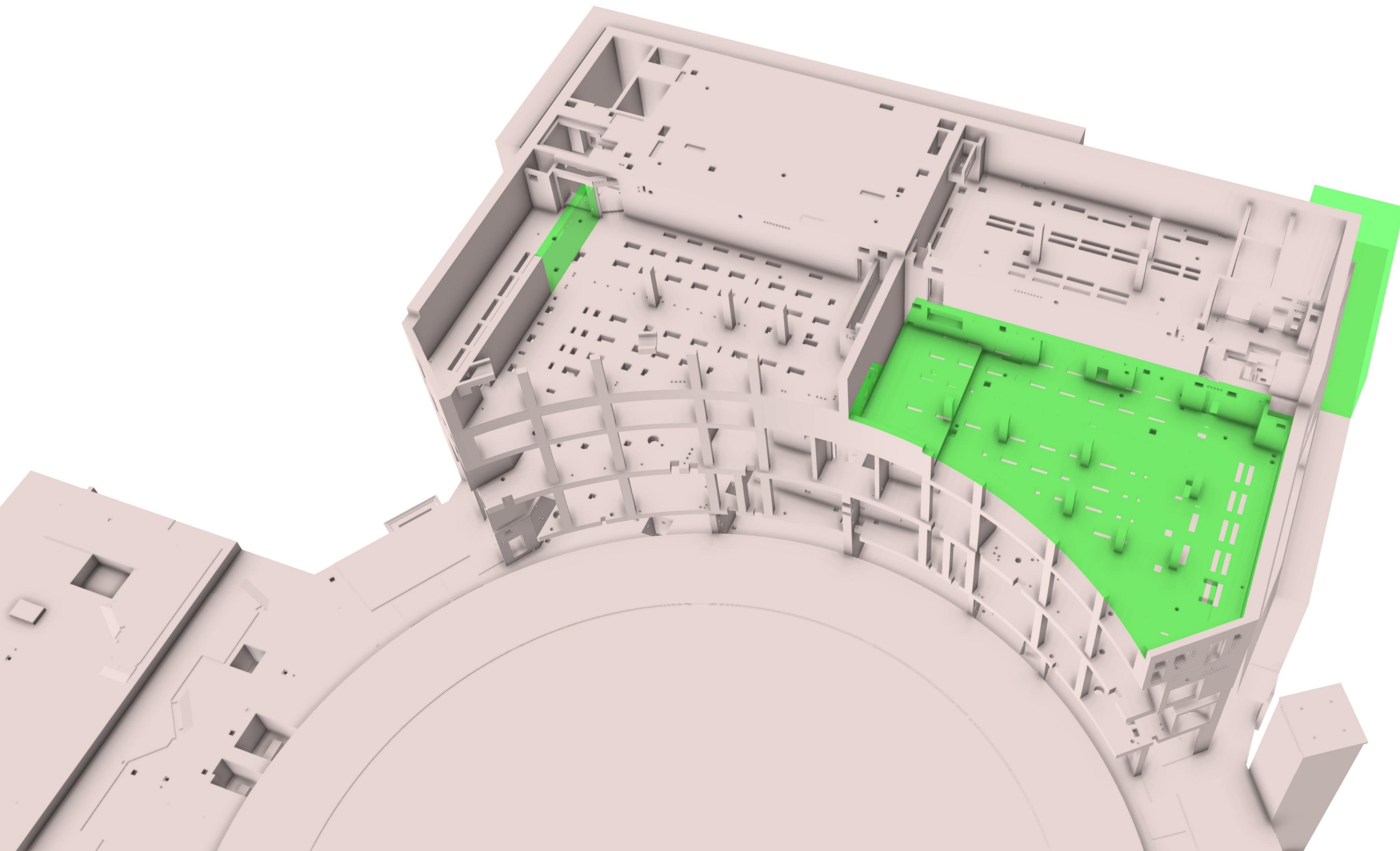
Ensure Scalability



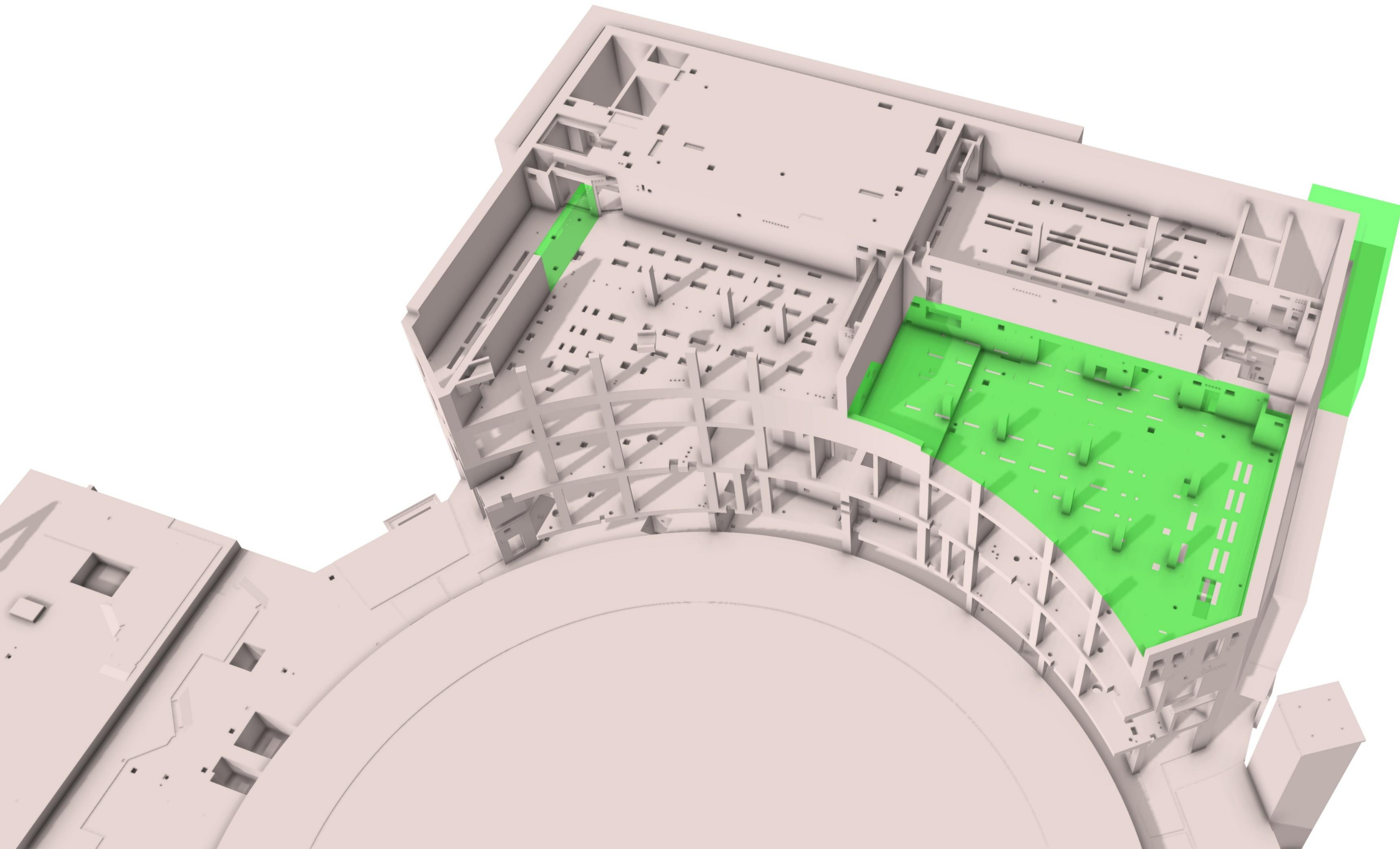
Average output **quality** out of the box



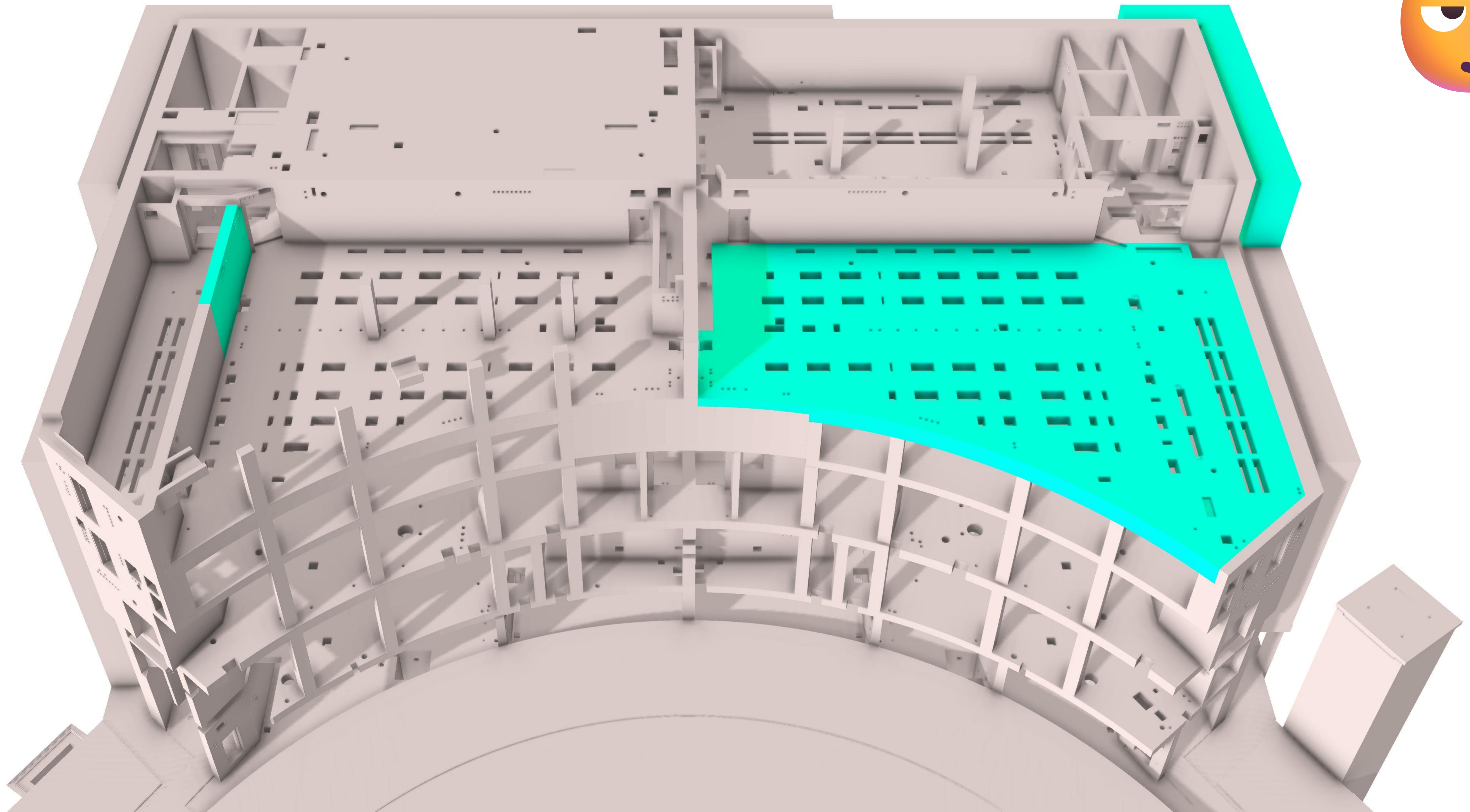
Let's turn on **Ambient Occlusion**



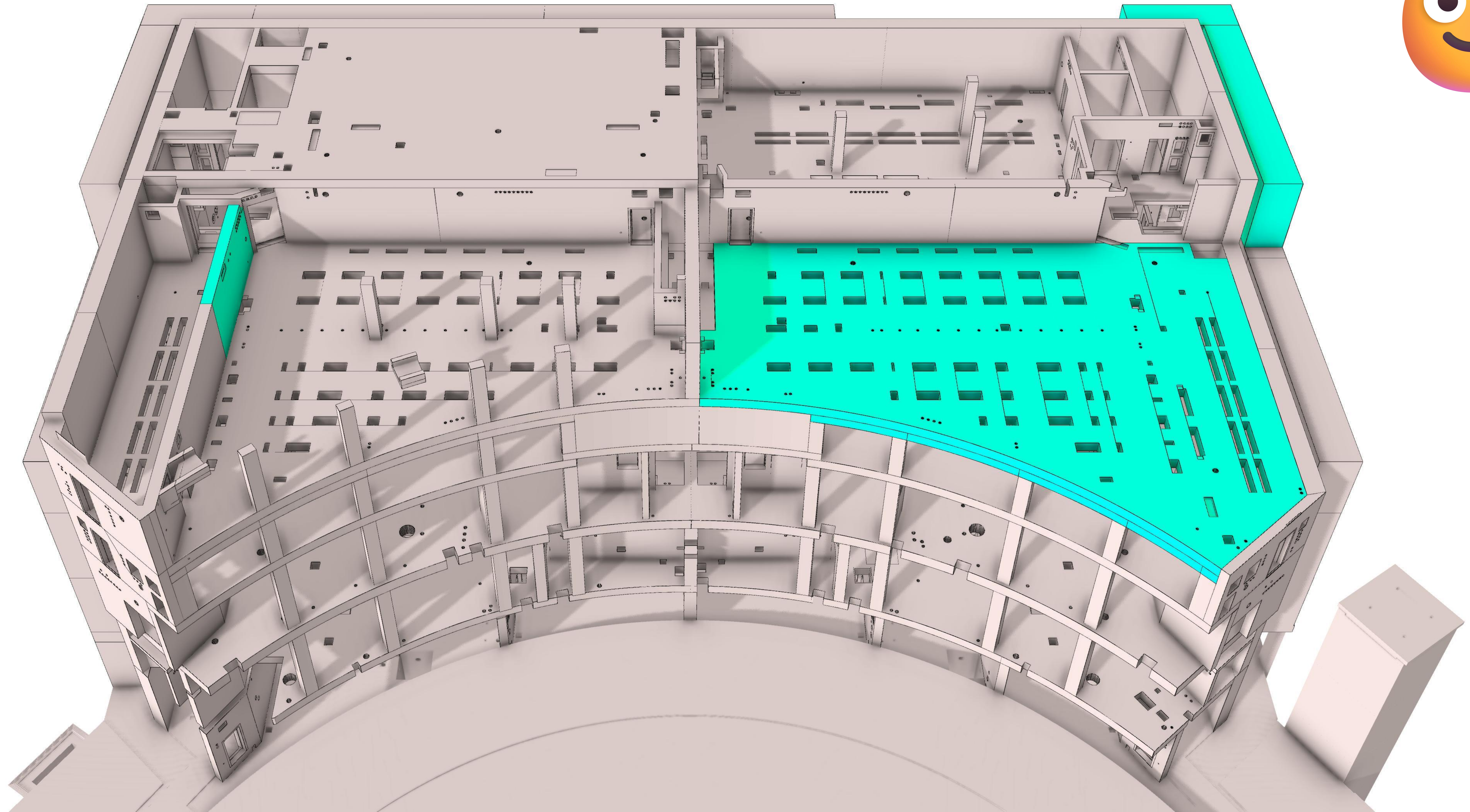
Let's turn on **Shadows**



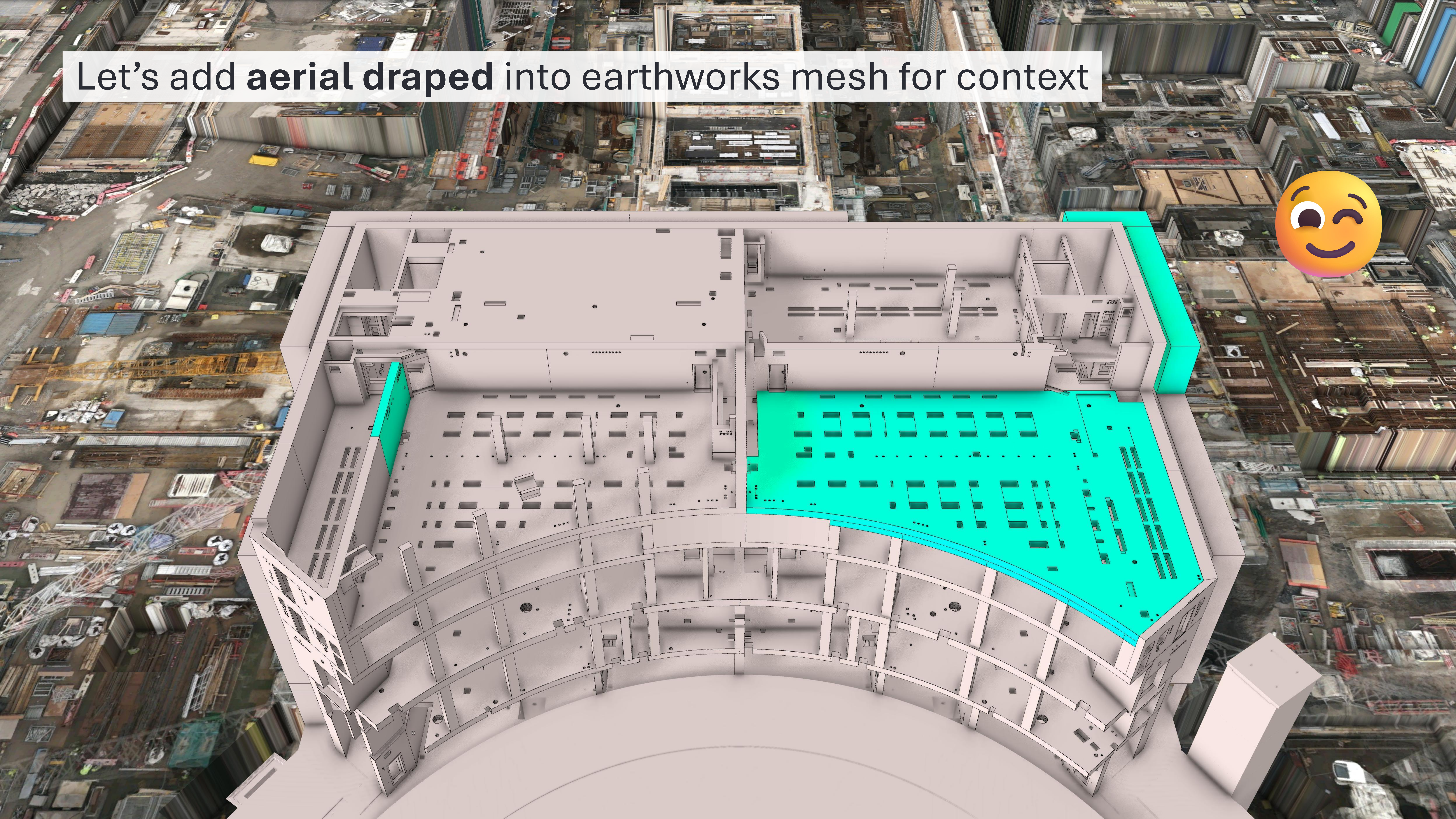
Standardise **Cameras** for consistency | Use custom appearance profiles



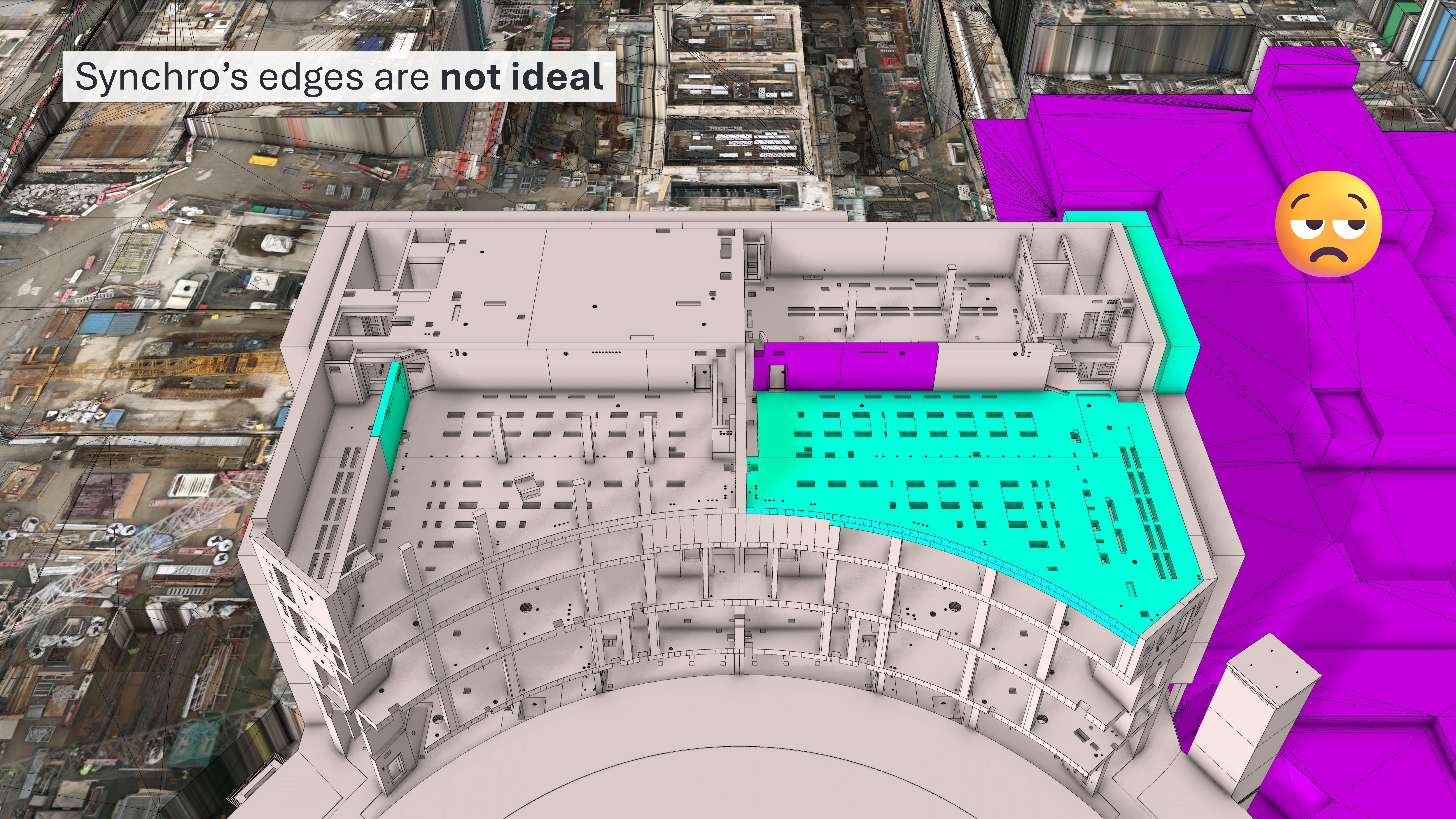
Let's add **contour lines** to see **construction joints**



Let's add **aerial draped** into earthworks mesh for context



Synchro's edges are **not ideal**



Let's add **object IDs** to help guys on site



647

621

605

## LOOKING BACK



## LOOKING FORWARD



## LOOKING FORWARD

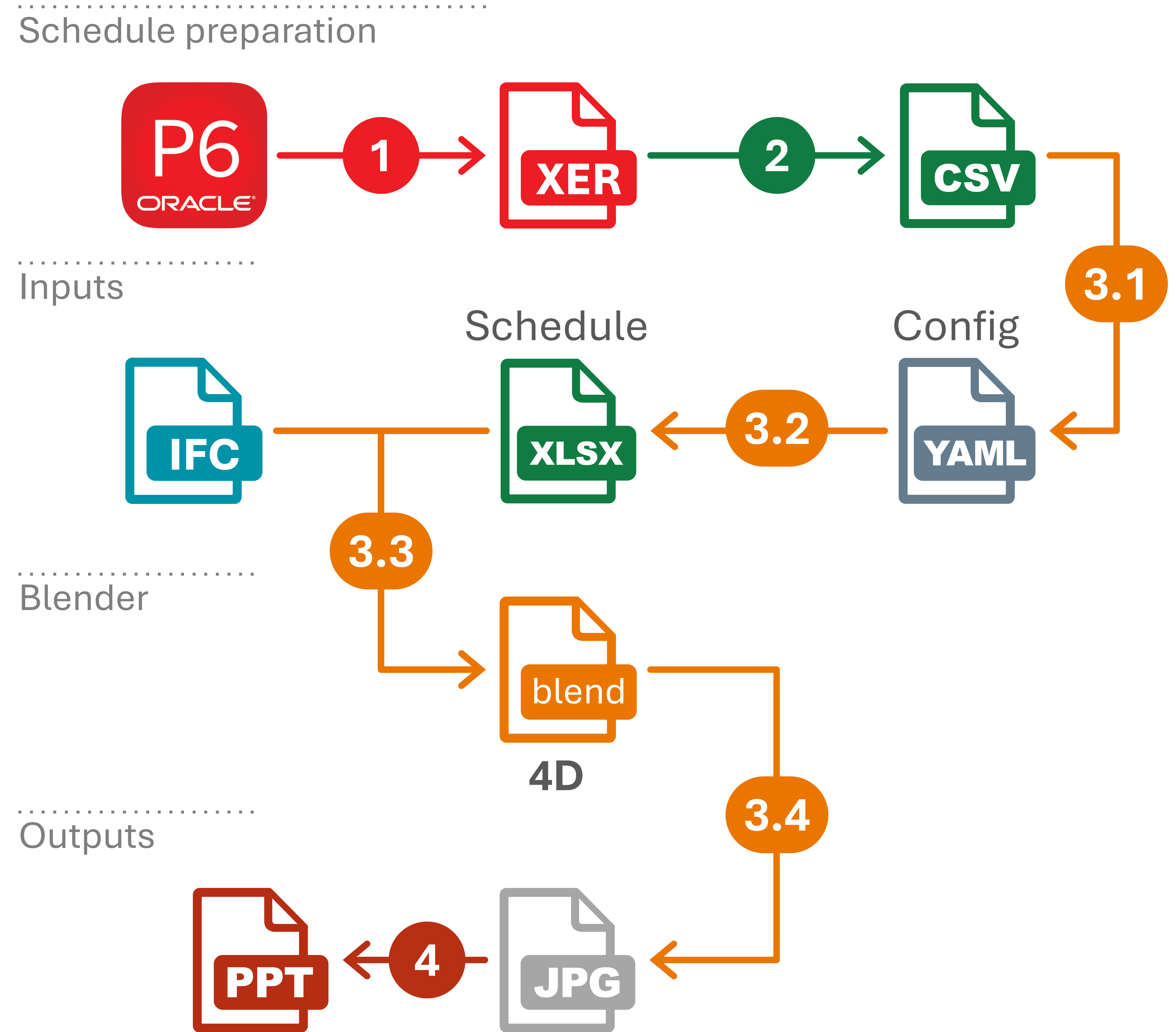


## LOOKING BACK



# 4D Pipeline v1 – Hinkley Point C

- 1 Manually **export schedule** from Primavera P6 to an XER file
- 2 **Schedule preparation** using Python script to:
  - 2.1. Convert XER tables into CSV files
  - 2.2. Merge necessary data into CSV per project
- 3 **Animation logic and Blender** using Python to:
  - 3.1. Read configuration YAML file
  - 3.2. Read CSV per project and prepare all 4D logic in Excel
  - 3.3. Blender file preparation
    - a) Import IFC
    - b) 4D Keyframing based on external Excel
  - 3.4. Export images
- 4 PowerPoint with macro imports all images



# 1 => N

## How we scale

- Simplification & Progress reporting.
- Replicate the complete environment to users.
- Centralized Channel for updates and feedbacks.
  - Git Repo for update distribution.
  - Group Chat for feedback and announcement

## Traceability

- Record Input Files and parameters. i.e. .blend file
- Logging for runtime parameter collection
- Able to replicate the runtime environment from collected data
- Remote debugging for detailed troubleshooting.

# 4D Pipeline v2 – Sizewell C pilot

1

Manually **export schedule** from Primavera P6 to an **XER** or **XML** file, schedules in **Excel** also work

2

**Blender file preparation** using Blender addon that uses Python to:

- 2.1. Read Configuration XLSX file
- 2.2. Import IFCs - one .blend per model, then append into a master .blend
- 2.3. Read schedules XER, XML or XLSX, parses and stores directly inside the .blend
  - a) Automatch tasks with objects
  - b) 4D keyframing based on embedded data

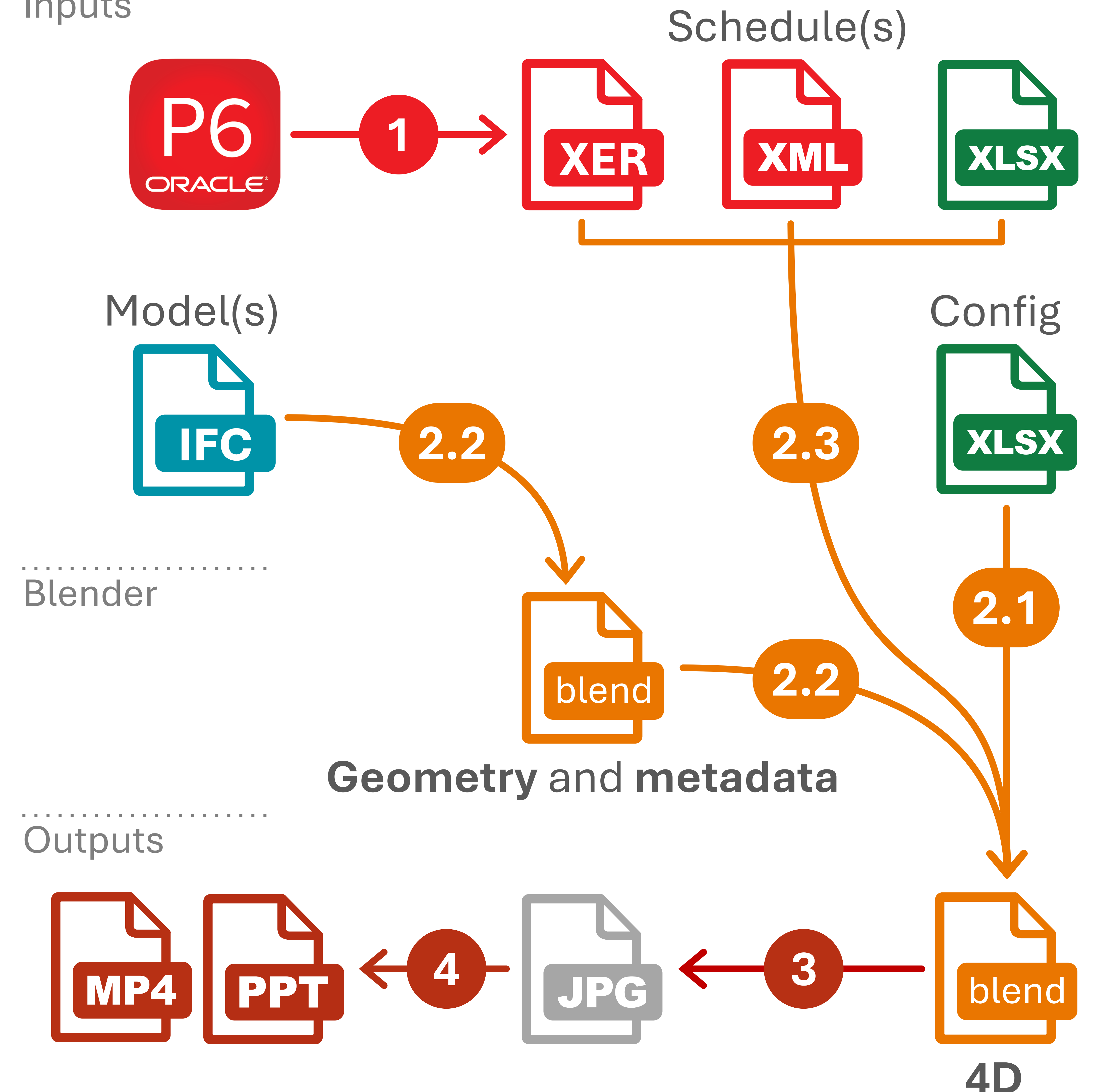
3

Export images

4

PowerPoint with macro imports all images, or turn the images into video with a Python script

Inputs



# 4D Pipelines comparison

|   |                  | v1                                                               | v2                                                                                                                                  |
|---|------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Execution        | Headless <b>batch</b> only                                       | Headless <b>batch</b> , or interactively as a Blender <b>add-on</b>                                                                 |
| 2 | Schedule inputs  | <b>XER</b> only                                                  | <b>XER</b> , <b>XML</b> and <b>XLSX</b>                                                                                             |
| 3 | Schedule storage | <b>Outside</b> Blender, in XLSX                                  | <b>Inside</b> Blender                                                                                                               |
| 4 | Model inputs     | <b>One IFC</b> per run                                           | <b>Multiple IFCs</b> , merged into one combined scene                                                                               |
| 5 | Object metadata  | Not cached, re-read from source each run                         | Cached as <i>[PropertySet]Name</i> custom properties                                                                                |
| 6 | Feature scope    | 4D sequencing for <b>reporting</b>                               | Same as v1 plus extra functionalities for <b>Planning</b> : colouring, storey explode, Multi-programme sync, Automatch, Gantt Chart |
| 7 | Animation speed  | Fixed per build: a new cadence needs regenerating and rebuilding | Change anytime in Blender: pick day/week/month/year, then re-animate                                                                |

Insight Plan

BIM

+

Scene

⚙️

📄

🕒

🔗

📹

👁️

🛠️

🕒 4D

Schedule Import

F:\K10 and Tower Cranes.xml

?

Not synced

F:\F06\_28JUN26.xer

?

Not synced

F:\B1 Backfill.xlsx

?

Not synced

F:\Granular Backfill.xlsx

?

Not synced

Not synced

↻ Sync All

🔗 Automatch All

Create Animation

Frames

1

1

Day

Start Date

2017-01-01

Finish Date

2028-05-02

🕒

Guess

Start Frame

1

Finish Frame

5000

📏

Calculate Frame Range

Speed Type

Frame-based

🎬

Create Animation

Jump to Date

🕒

Current: 2023-05-04 (frame 2796)

⏮️

⏭️

▶️

Jump

> ☒ Timeline Bar

> ☐ Timeline HUD

Add from Clipboard/Text

Enter P6 XML/XER file paths (one per line):

"F:\K10 and Tower Cranes.xml"

"F:\F06\_28JUN26.xer"

"F:\B1 Backfill.xlsx"

"F:\Granular Backfill.xlsx"

"F:\K10 and Tower Cranes.xml""F:\F06\_28JUN26.xer""F:\B1 Backfill.xlsx""F:\Granular Backfill.xlsx"

Tip: Copy paths from Windows Explorer or a text file

OK

Cancel

Day

Real Duration Unit

Day

Week

Month

Year

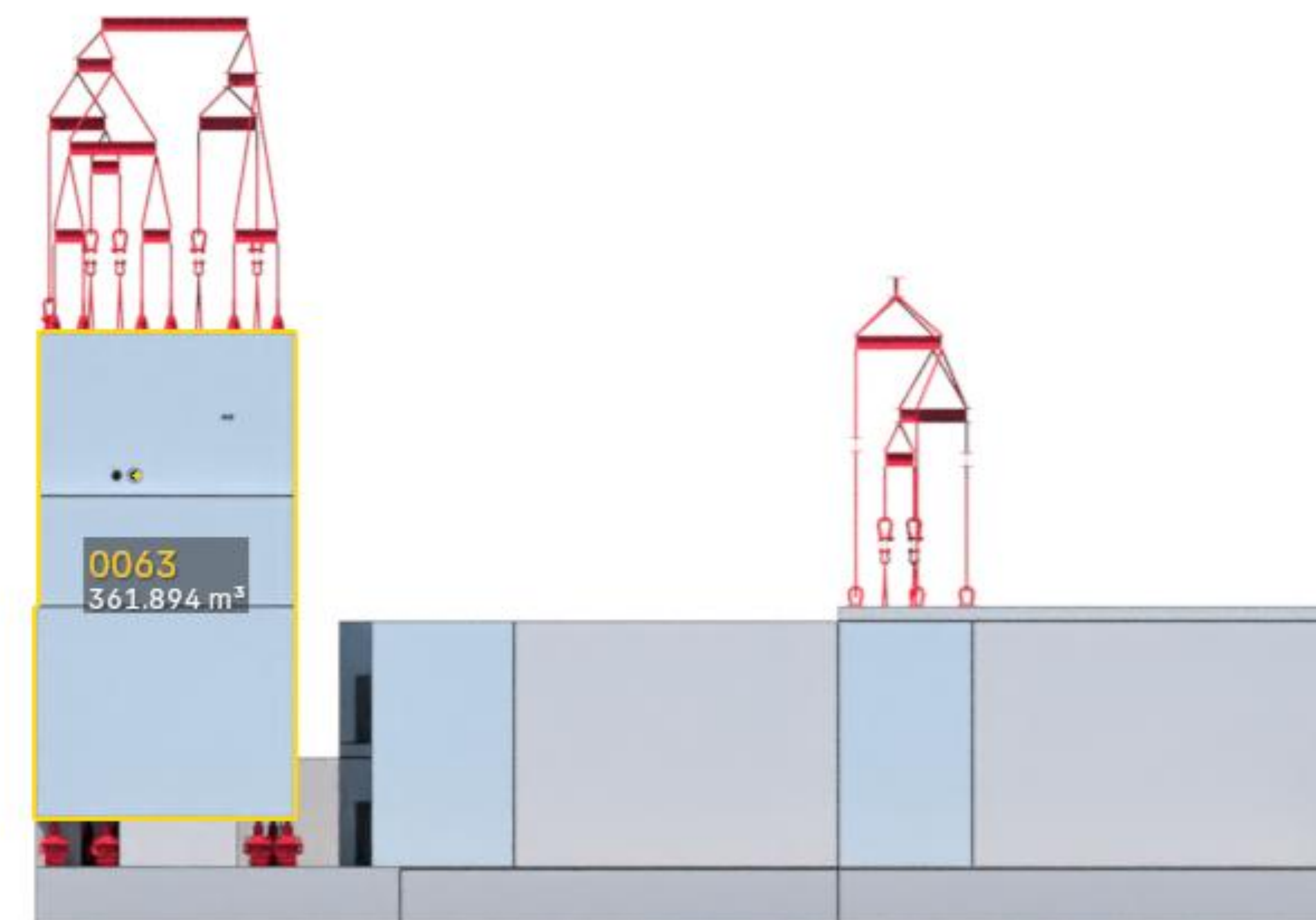
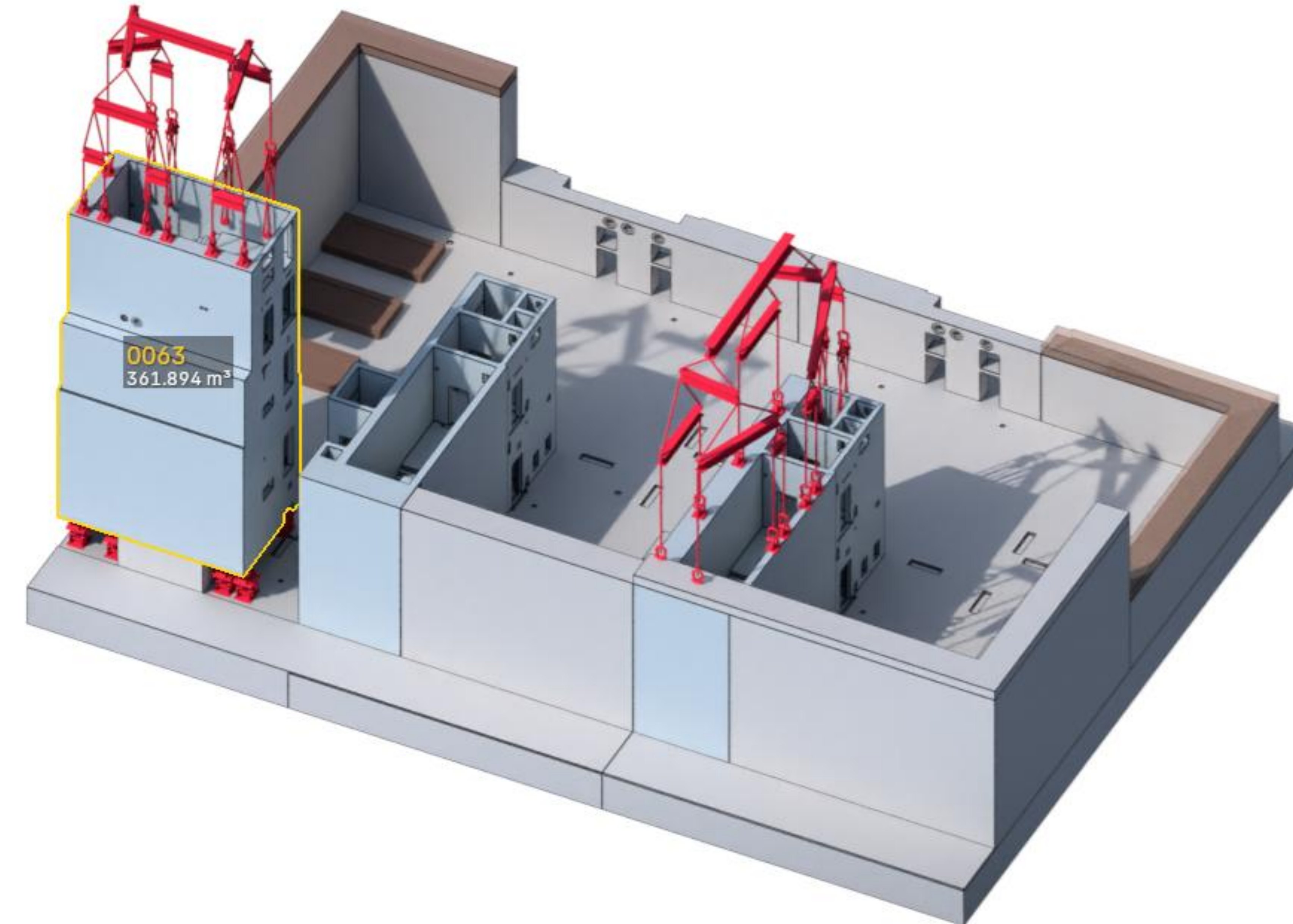
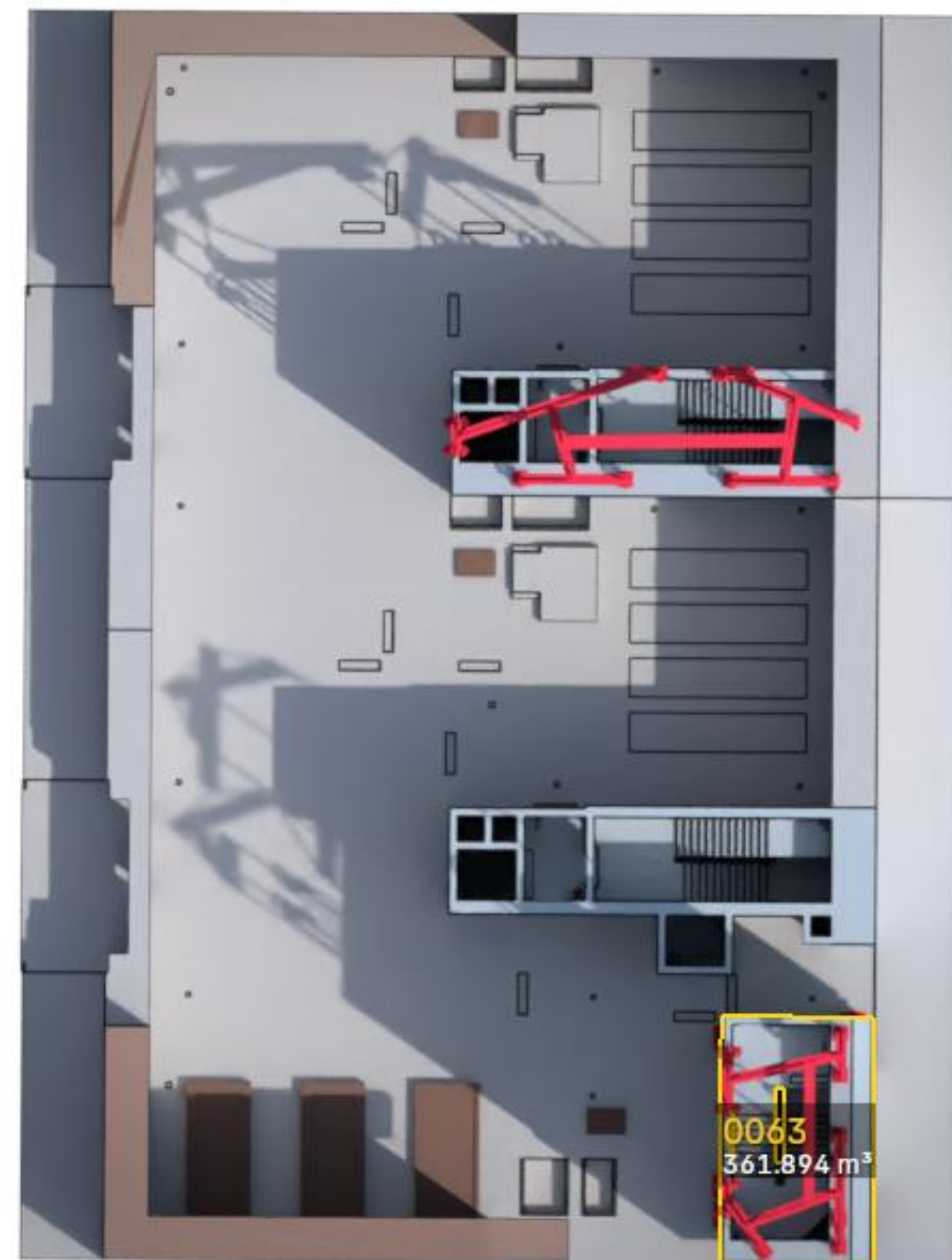
# 4D Pipeline v2 – New Visual Capabilities

## Presentation

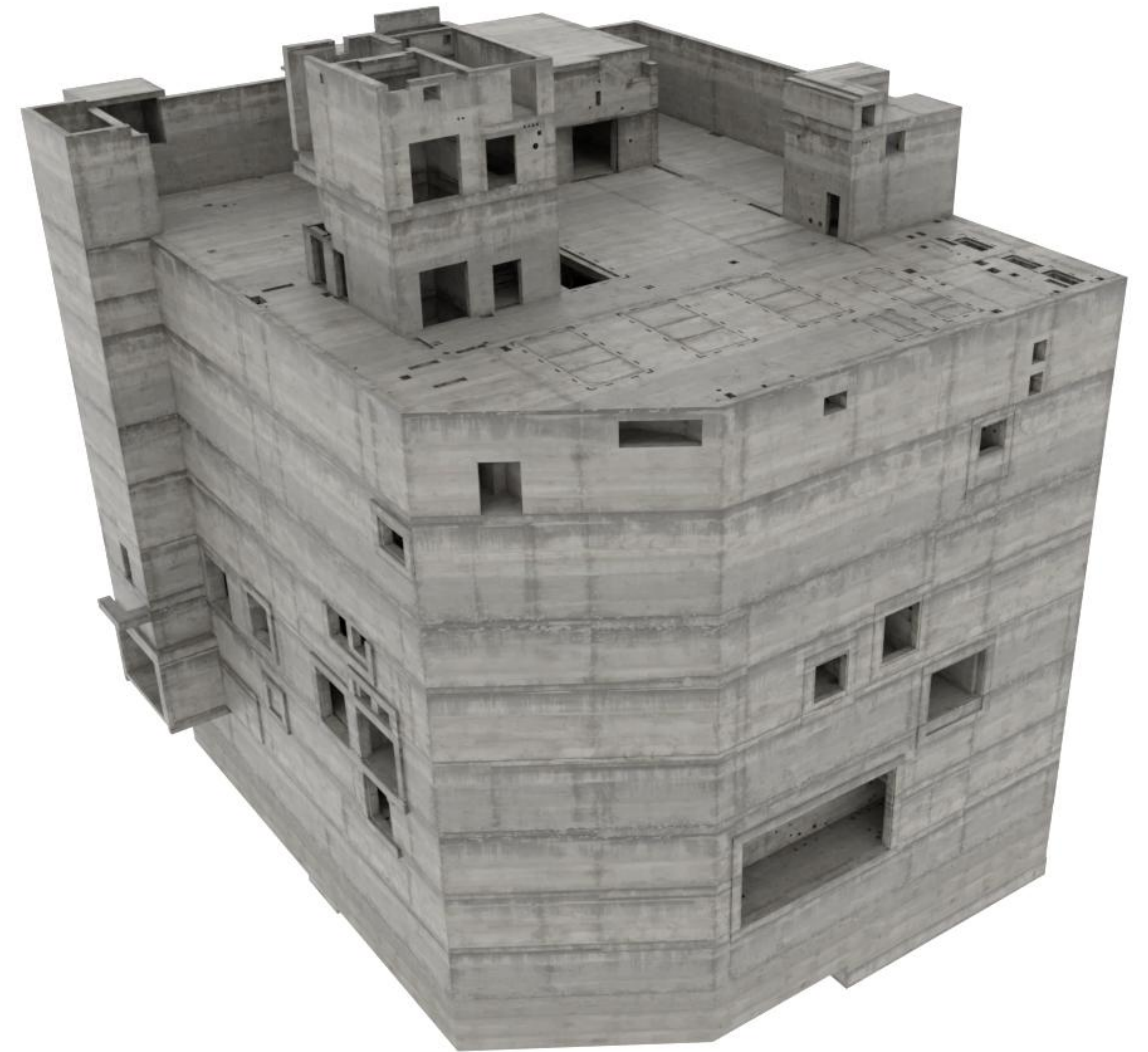
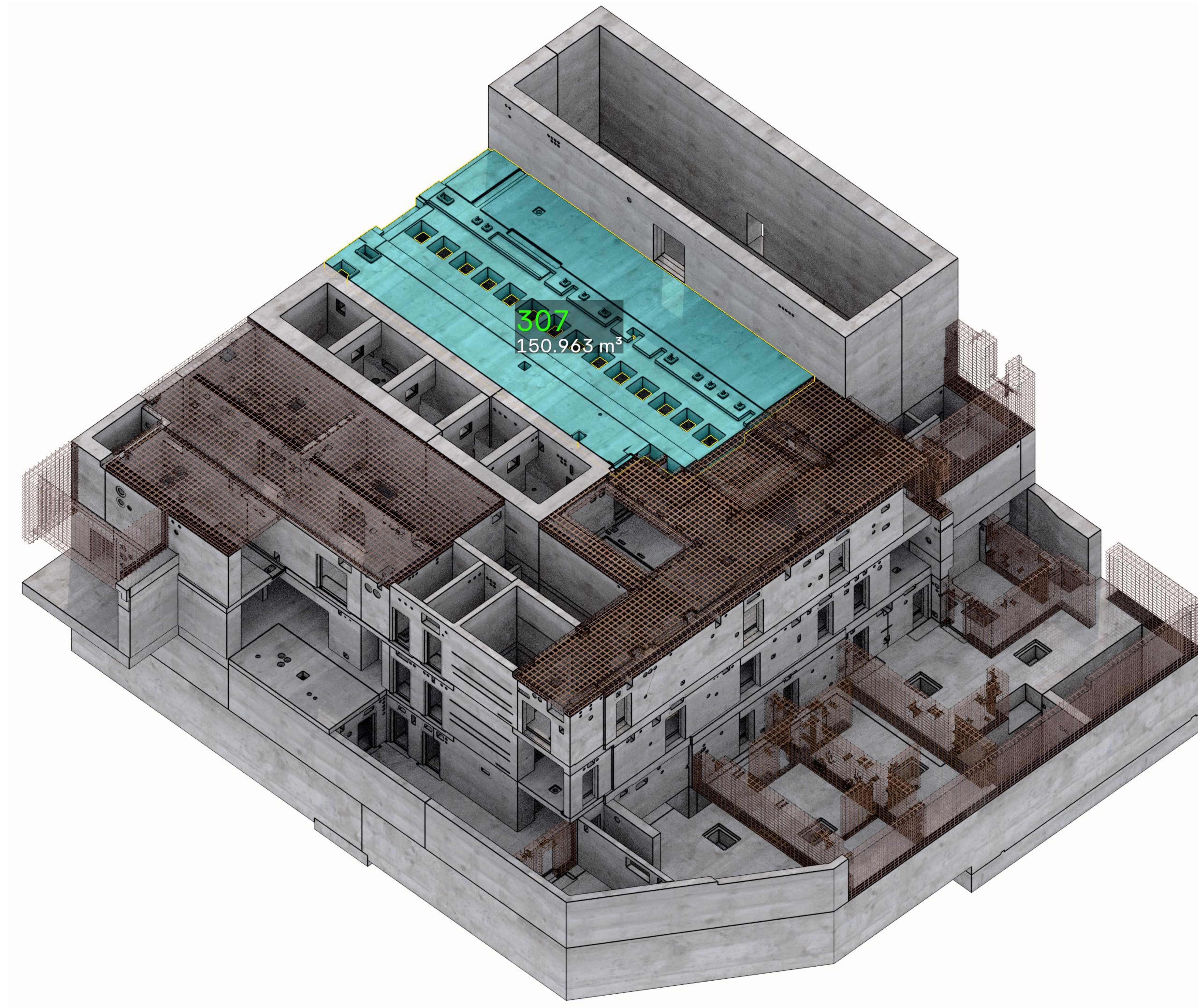
- Animated camera from close-up to full building
- Multi-view render: 1, 2, 3 or 4 views
- Custom labels on active tasks
- Growth effect (e.g. dropping precast, left to right on a slab)
- Animated Monthly aerial

## Model content

- Realistic materials
- Temporary works and secondary steel
- Neighbouring buildings, backfill and site logistics



# 4D Pipeline v2 – Realistic materials



# Key Takeaways

- ✓ **Blender sits in the right place for what we need now**  
Scheduled 4D image sequences, on modest hardware, easy to deploy and a Python/headless workflow we already know how to automate and customise.
- ✓ **Automation demands discipline**  
Models and P6 schedules must match precisely and remain accurate.
- ✓ **Python + IfcOpenShell allow manipulation of IFC models**  
Fix common model issues programmatically with Python before 4D sequencing.
- ✓ **Multiprocessing cuts run times**  
Process multiple IFC/Blender files, or renderings in parallel.
- ✓ **Faster workflows = time for more creative & precise outputs**  
Cutting times dramatically opens space to experiment with alternatives, improve accuracy, and create richer, more decision-useful 4D models.

